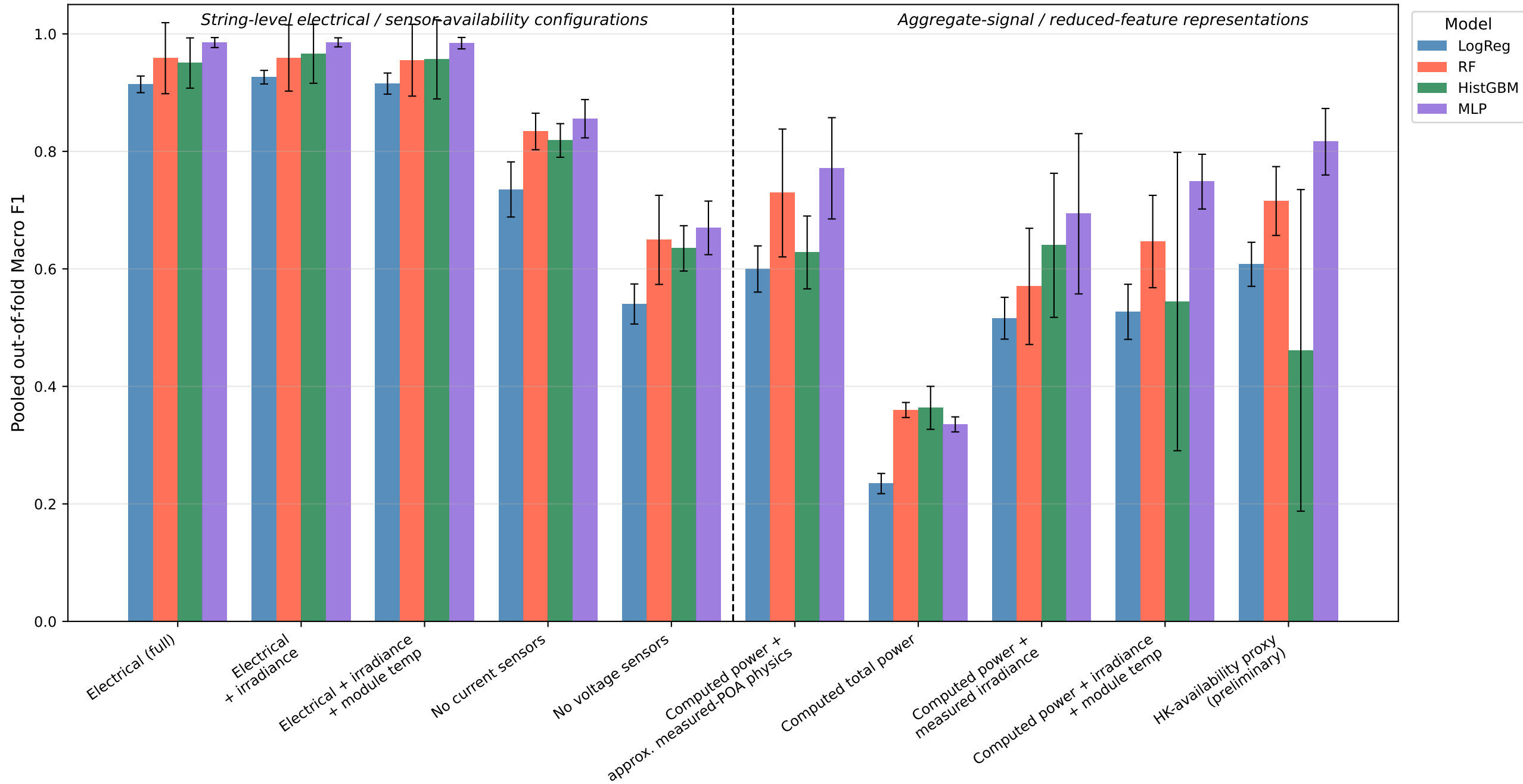


Objective 1 — group-disjoint results: 5-fold CV, all 16 recording days



Error bars show fold standard deviation (ddof=1, n=5). Aggregate-signal sets require the same string sensors as the full electrical set — $p_{dc_computed} = v_{dc1} \times i_{dc1} + v_{dc2} \times i_{dc2}$. NO_CURRENT and NO_VOLTAGE are the genuine hardware-removal ablations.